User Study – ML2++ (Textual DSL)

Instructions: Fill Section A once. For Section B, answer for *each* use case you worked on. If you did multiple use cases, **duplicate Section B** per use case.

Participant ID: ______________________ **Use case(s):** ☑ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

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Section A – Background (answer once)

This section helps us understand your experience level so we can analyze results fairly.

A1. Educational level *(highest completed)*

Tell us the highest level of education you have finished.

☐ High School ☐ Vocational/Technical ☐ Bachelor's ☑ Master's ☐ Doctorate ☐ Other: ______________________

A2. Primary role *(pick the closest)*

Choose the option that best matches your current job or main activity. If none fits, select "Other".

☐ IoT/CPS Developer ☐ MDE Practitioner ☐ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☑ Software Architect / Full-stack ☐ System Integrator ☐ Early Adopter / Innovator ☐ Other: Devops

A3. Self-rated expertise *(tick one per row)*

For each area, select your skill level. "IoT/CPS" = connected devices/systems. "MDE" = using models to guide or generate code. "ML/Data Science" = working with data and ML.

	Beginner	Intermediate	Advanced	Expert
IoT/CPS development	☑	☐	☐	☐
Model-Driven Engineering (MDE)	☑	☐	☐	☐
Machine Learning / Data Science	☐	☑	☐	☐

A4. Prior experience *(used DSLs/modeling tools/ML frameworks before?)*

Have you used similar tools (e.g., modeling editors, DSLs, ML libraries/frameworks)? tensorflow

☑ Yes ☐ No

Section B – Task-Specific (duplicate this page *per use case*)

This B-section is for: ☑ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

If you did more than one use case, copy this whole Section B for each one and fill it separately.

B1. Time & Effort

Please record your times honestly. If unsure, give your best estimate. "Extra learning time" = time reading docs/tutorials to figure things out.

B1. Time spent (minutes)

Minutes you spent defining/editing the DSL, generating code, reviewing outputs, and any extra learning time.

	ML2++ (Textual DSL)
Defining DSL + generate + review	60min
Extra learning time	45min

B2. Statements written (DSL) *(exclude blanks/comments)*

Approximate number of DSL lines/statements you actually typed or edited.

	ML2++ (Textual DSL)
ML2++ DSL statements	156

B2. Task Completion & Ease of Use

Tell us if you finished the task and how easy/hard each step felt. Then write how many minutes you spent per step.

B3. Were you able to finish the use case?

"None" = could not complete; "Partial" = finished some parts; "Complete" = finished everything.

☐ None ☐ Partial ☑ Complete

B4. Ease by workflow step (with per-step time)

Rate difficulty (1=very easy, 5=very difficult) and write minutes spent. "Model configuration" = choosing algorithm and hyperparameters. "Model evaluation" = checking results with plots/metrics.

Step	1	2	3	4	5	**Time**
Whole workflow (end-to-end)	☐	☐	☑	☐	☐	___
Data preprocessing	☐	☐	☑	☐	☐	___
Model configuration (algorithm, hyperparameters)	☐	☑	☐	☐	☐	___
Model training	☐	☑	☐	☐	☐	___
Model evaluation (metrics/plots/inspection)	☑	☐	☐	☐	☐	___

Tip: use a simple stopwatch and pause for breaks. If unsure, estimate.

B5. Ease of ML2++ elements *(1=Very easy → 5=Very difficult)*

Rate how easy it was to work with each part of the language/editor.

Element	1	2	3	4
Data types & objects	☐	☑	☐	☐
Things & behaviours (statecharts)	☐	☑	☐	☐
Ports & properties	☐	☑	☐	☐
Model configuration options (algorithm/hyperparameters)	☐	☑	☐	☐
Configuration (instances/connectors)	☐	☑	☐	☐

B3. Learning & Usability

These questions focus on how quickly you learned the tool and how natural it felt.

B6. Learning curve *(1=Very easy → 5=Very difficult)*

How much effort did it take to learn enough to finish this use case?

☐1 ☐2 ☑3 ☐4 ☐5

B7. Intuitiveness *(1=Very intuitive → 5=Not intuitive)*

Did the tool feel natural/obvious to use, or did it feel confusing?

☐1 ☐2 ☑3 ☐4 ☐5

B8. Hardest part to learn *(briefly)*

Name the single most confusing or time-consuming thing to learn.

learn how it work

B4. Debugging & Error Handling

Tell us about problems you faced and how hard they were to fix.

B9. Error frequency *(0=None, 1=Few, 2=Moderate, 3=Many)*

How many problems blocked you (errors, failed runs, missing packages, wrong paths, etc.)?

☐0 ☐1 ☑2 ☐3

B10. Clarity of error messages *(1=Not clear → 5=Very clear)*

When something went wrong, did the messages help you understand the fix?

☐1 ☑2 ☐3 ☐4 ☐5

B11. Debugging effort *(1=Easy → 5=Hard)*

Overall, how hard was it to fix your issues?

☐1 ☑2 ☐3 ☐4 ☐5

B12. Debugging time (minutes) *(estimate)*

About how many minutes did you spend fixing problems (total for this use case)?

20 min

B5. Code Quality, Maintainability, Flexibility

These questions refer to the generated *code you viewed and how easy it would be to change things later.*

B13. Generated code quality *(1=Poor → 5=Excellent)*

How good is the generated code (readable, well-structured, correct)?

☐1 ☐2 ☐3 ☑4 ☐5

B14. Maintainability *(1=Very difficult → 5=Very easy)*

If you had to update/change your solution later (e.g., new dataset, new horizon), how easy would it be?

☐1 ☐2 ☑3 ☐4 ☐5

B15. Flexibility / Customisation *(1=None → 5=Very flexible)*

How easily can you change models, hyperparameters, thresholds, or add steps without workarounds?

☐1 ☐2 ☐3 ☑4 ☐5

Section C – Overall Assessment (ML2++ only)

Your overall opinion and which features helped you most.

C1. Overall satisfaction *(1=Very dissatisfied → 5=Very satisfied)*

Your overall feeling about using ML2++ for this use case.

☐1 ☐2 ☐3 ☐4 ☑5

C2. Would you use ML2++ for your next similar project?

If you had a similar project tomorrow, would you choose this approach? Tell us briefly why.

☑ Yes ☐ No ☐ Undecided *Why?* beacuse you can focus on your problem and design our model better

C3. Feature effectiveness *(1=Ineffective → 5=Very effective)*

How useful were these features for you?

Feature	1	2	3	4	5
Time-series forecasting models	☐	☐	☐	☑	☐
Automatic preprocessing & plots	☐	☐	☑	☐	☐
Training configuration options	☐	☐	☑	☐	☐
Multi-day forecasting accuracy	☐	☐	☑	☐	☐

C4. Favourite feature: define our models

What was the single most helpful feature?

C5. Least favourite feature: preprocessing

Which feature felt least helpful or got in your way?

C6. Missing features you would like: design to have better syntax i think yaml file its good

What do you wish the tool had (e.g., a plot, algorithm, wizard, or shortcut)?

Section D – Open Feedback

Tell us anything else that would help us improve.

D1. Barriers to adoption *(what would stop you from using ML2++ in production?)*

Examples: setup, missing plots, docs, integration, performance, governance.

low usage on community and lack of documentaction

D2. One improvement you would prioritize

If we could fix/add only one thing, what would help you the most?

support modular
use configuration file like yaml
try lower conf code

Scale anchors used throughout for consistency:

- Ease / Learning: 1=Very easy, 2=Easy, 3=Moderate, 4=Difficult, 5=Very difficult.
- Intuitiveness: 1=Very intuitive, 2=Intuitive, 3=Neutral, 4=Somewhat confusing, 5=Not intuitive.
- Error clarity: 1=Not clear, 2=Somewhat clear, 3=Clear, 4=Very clear, 5=Extremely clear.
- Code quality: 1=Poor, 2=Fair, 3=Good, 4=Very good, 5=Excellent.
- Maintainability: 1=Very difficult, 2=Difficult, 3=Moderate, 4=Easy, 5=Very easy.
- Flexibility: 1=None, 2=Limited, 3=Some, 4=Flexible, 5=Very flexible.
- Feature effectiveness: 1=Ineffective, 2=Low, 3=Moderate, 4=Effective, 5=Very effective.

User Study – Manual Python Workflow (Method 1 Only)

Instructions: Fill Section A once. For Section B, answer *per use-case*. Duplicate Section B if you completed multiple use-cases.

Participant ID: ________________ **Your assignment:** ☐ **Method 1** (Manual Python)
Use case(s) you worked on: ☑ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

Section A – Background (answer once)

A1. Educational level *(highest completed)*
☐ High School ☐ Vocational/Technical ☐ Bachelor's ☑ Master's ☐ Doctorate ☐ Other: ________________

A2. Primary role *(pick the closest)*
☐ IoT/CPS Developer ☐ Data Scientist / ML Engineer ☐ Embedded SW Engineer ☐ Educator / Researcher ☑ Software Architect / Full-stack ☐ System Integrator ☑ Other: ________________ Devops

A3. Self-rated expertise *(tick one per row)*

	Beginner	Intermediate	Advanced	Expert
Python programming	☐	☐	☑	☐
Time-series / ML	☐	☑	☐	☐
Domain knowledge (your use-case)	☐	☑	☐	☐

A4. Prior experience *(used ML frameworks/libraries before?)* ☑ Yes ☐ No

Section B – Task-Specific (duplicate this page *per use-case*)

This B-section is for: ☑ U1 River Flow ☐ U2 Smart-Home Energy ☐ U3 Solar Energy

B1. Time & Effort

Record your own times. Estimate if needed. "Extra learning time" = time spent reading docs/tutorials.

B1. Time spent (minutes)

	Method 1 (Manual Python)
Writing/testing code (end-to-end)	14
Extra learning time	6

B2. Lines of code (LoC) *(exclude blanks/comments)*
Manual Python LoC: 108

B2. Task Completion & Ease of Use

Completion: None = could not finish; Partial = finished some parts; Complete = all parts.

B3. Were you able to finish the use-case with Manual Python?
Method 1 (Manual Python): ☐ None ☐ Partial ☑ Complete

B4. Ease by workflow step (Manual Python) *(1=Very easy, 5=Very difficult; tick one per cell)*

Step	1	2	3	4	5
Whole workflow	☐	☑	☐	☐	☐
Data preprocessing	☐	☐	☑	☐	☐
Feature engineering (lags/rolling)	☐	☑	☐	☐	☐
Model training	☑	☐	☐	☐	☐
Model evaluation/plots	☑	☐	☐	☐	☐

B1. Which steps did you complete? *(check all that apply)*

☑ **Preprocessing**
☑ **Model configuration**
☑ **Training**
☑ **Evaluation**
☑ **Plotting/Diagnostics**

B2. How difficult was each step? *(1=Very easy, 5=Very difficult; tick one per row)*

Step	1	2	3	4	5
Preprocessing	☐	☐	☑	☐	☐
Model configuration	☑	☐	☐	☐	☐
Training	☑	☐	☐	☐	☐
Evaluation	☑	☐	☐	☐	☐
Plotting/Diagnostics	☑	☐	☐	☐	☐

B3. Notes (issues, decisions, or gaps to revisit):

____

____

B5. Time spent per step *(record for this use-case; minutes or start/end)*

Step	**Start (hh:mm)**	**End (hh:mm)**	**Minutes**
Preprocessing	4	00	00
Model configuration	1	00	00
Training	1	00	00
Evaluation (metrics/baseline)	1	30	00
Plotting / Diagnostics	1	30	00
Debugging (errors, crashes, outputs)	5	00	00
Extra learning / searching / docs	6		
Total			20

Notes: "Learning" includes reading docs, web searches, tutorials, setup notes, and guides. "Debugging" covers error fixing, crashes, failed runs, and investigating unexpected results.

B6. Comments (bugs, friction points, ideas):

LSTM train: 275s Test: 4s
GRU train: 263s Test: 3s

Section E – IoT Use Case Template Reuse (Part 2)

Instructions: This section evaluates your experience reusing the provided IoT pipeline template (e.g., River Flow Forecasting) and adapting it for your own use case. Please answer honestly.

E1. Template Understanding
How clear was the IoT pipeline template (overview, scripts, execution order)?
❑ 1 Very unclear ☑ 2 Unclear ❑ 3 Neutral ❑ 4 Clear ❑ 5 Very clear

E2. Adaptation Effort
How easy was it to adapt the template to your chosen use case?
❑ 1 Very difficult ☑ 2 Difficult ❑ 3 Neutral ❑ 4 Easy ❑ 5 Very easy

E3. Reuse of Offline Training Code
How straightforward was it to reuse the provided offline training code with your dataset?
❑ 1 Not at all ❑ 2 A little ☑ 3 Neutral ❑ 4 Mostly ❑ 5 Very straightforward

E4. Time & Effort (minutes)
Understanding the template: 7
Adapting scripts (sensors, gateway, analytics): 5
Running end-to-end pipeline: 60

E5. Problems & Debugging
Did you face errors while reusing the template? ☑ Yes ❑ No

If yes, briefly describe the main issues: package installing

How difficult were they to fix?
❑ 1 Very easy ❑ 2 Easy ☑ 3 Neutral ❑ 4 Hard ❑ 5 Very hard

E6. Usefulness
How useful was the template for accelerating your implementation?
❑ 1 Not useful ❑ 2 Slightly useful ❑ 3 Neutral ☑ 4 Useful ❑ 5 Very useful

E7. Suggested Improvements
What changes would make the IoT template easier to reuse for future use cases?

combine server and analytics and scale data i train give more data for traing

Scale anchors (for consistency):

- Clarity / Ease: 1 = Very unclear/difficult, 2 = Unclear/difficult, 3 = Neutral, 4 = Clear/easy, 5 = Very clear/easy. 3
- Debugging difficulty: 1 = Very easy, 5 = Very hard. 3
- Usefulness: 1 = Not useful, 5 = Very useful. 5